A literature answer to the general surface-area-measure question for log-concave functions

Status note for arXiv:2006.16933

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Status

The open problem in L. Rotem, *Surface area measures of log-concave functions*, arXiv:2006.16933, is answered by the same author's later paper *The anisotropic total variation and surface area measures*, arXiv:2206.13146. The answer is Definition 1.2, Definition 1.4, and Theorem 1.5 of the later paper.

The source question

For an integrable upper-semicontinuous log-concave function $f = e^{-\varphi}$ on $\mathbb{R}^n$, arXiv:2006.16933 uses the push-forward

$$S_f = (\nabla\varphi)_\#(f\,dx).$$

This correctly gives the interior contribution for sufficiently regular functions, but if $f = \mathbf{1}_K$ is the indicator of a convex body, it reduces to $|K|\delta_0$ and loses the classical surface-area measure S_K. Accordingly, PDF page 5 asks for an extension of its Definition 1.4 and first-variation Theorem 1.5 that works for every log-concave f, including both smooth functions and indicators of convex bodies.

The later answer

Let $K_f = \overline{\{x : f(x) > 0\}}$. ArXiv:2206.13146, Definition 1.2, associates to $f = e^{-\varphi}$ two measures:

$$\mu_f = (\nabla\varphi)_\#(f\,dx) \quad \text{on } \mathbb{R}^n, \qquad \nu_f = (n_{K_f})_\#\left(f\,d\mathcal{H}^{n-1}\big|_{\partial K_f}\right) \quad \text{on } S^{n-1}.$$

Here n_{K_f} is the almost-everywhere defined outer Gauss map. Definition 1.4 calls the pair (μ_f, ν_f) the surface-area measures of f.

Theorem 1.5 on supporting PDF page 3 proves, for arbitrary $f, g \in \mathrm{LC}_n$ with $0 < \int f < \infty$, the unrestricted formula

$$\delta(f,g) = \int_{\mathbb{R}^n} h_g\,d\mu_f + \int_{S^{n-1}} h_{K_g}\,d\nu_f,$$

with no smoothness or strict-convexity assumptions. The paper further explains on PDF pages 4–5 that the indicator perturbation $g = \mathbf{1}_L$ is an anisotropic coarea formula.

For $f = \mathbf{1}_K$, the definitions give

$$\mu_f = |K|\delta_0, \qquad \nu_f = S_K.$$

Thus the second term restores exactly the boundary contribution missing from the one-measure definition, while for full-support smooth functions the boundary is empty and $\nu_f = 0$. This is precisely the requested common extension.

Conclusion

The question is not a viable new-proof target: arXiv:2206.13146 supplies the general definition and the complete first-variation theorem, with the indicator-of-a-convex-body case built into the same framework.

References

L. Rotem, *Surface area measures of log-concave functions*, arXiv:2006.16933 (2020), especially PDF page 5.

L. Rotem, *The anisotropic total variation and surface area measures*, arXiv:2206.13146 (2022), Definitions 1.2 and 1.4 and Theorem 1.5, PDF pages 2–5.